

SASTRA DEEMED UNIVERSITY
(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

May 2024

Course Code: CHY101

Course: ENGINEERING CHEMISTRY

QP No. :U173-2

Duration: 3 hours

Max. Marks:100

PART – A

Answer all the questions

10 x 2 = 20 Marks

1. Why do we need to add more soap in hard water for lather formation?
2. Two sample waters A and B were taken from Cauvery and Mahananda rivers respectively. For 20 ml sample A 15 ml 0.1 M EDTA solution was consumed to obtain the end point, whereas, for 10 ml sample B 7.5 ml 0.2 M EDTA was consumed. Comment on the relative hardness of sample waters A and B.
3. Why do we use ammonia buffer during estimation of hardness by EDTA?
4. What are the advantages of electroless plating over electroplating?
5. The rate of corrosion of Al is lower than Fe, although the SRP for Al and Fe are -1.66 V and -0.44 V respectively. Justify the observation.
6. Fe What is the role of Pickling and etching in electroplating?
7. A water droplet always takes round shape on hydrophobic surface - provide a reason of this observation.

8. What is aquadag? Mention their use.
9. Mention the importance of dispersed phase in a given matrix.
10. Mention two uses of nanomaterial in healthcare.

PART - B

Answer all the questions

4 x 15 = 60 Marks

11. (a) Compare phosphate and calgon conditioning methods. (6)
- (b) 200 ml of a sample of water required 5 ml of (N/20) sodium carbonate solution for complete precipitation of calcium ion (Ca^{2+}) and calcium carbonate. Calculate the hardness in ppm. (4)
- (c) The total hardness of 1,000 litres of water was completely removed by a zeolite softener. The zeolite softener required 30 litres of NaCl solution, containing 15 g/l of NaCl for regeneration. Calculate the hardness of water. (5)

(OR)

12. (a) Compare Zeolite process with Ion exchange method. (10)
 - (b) How can we differentiate COD from BOD? A water sample taken from our boy's hostel was analyzed for BOD and COD. Comment on their relative value in ppm unit. (5)
 13. (a) How the potential develops in an electrode when it kept in its own solution? Is it safe to keep AgCl solution in Copper container? Provide your reason in favor of your answer. (8)
 - (b) Calculate the cell potential and write the cell reaction for the given cell. Calculate the equilibrium constant and change of free energy for the same. (7)
- $\text{Zn(s)}/\text{Zn}^{2+}(\text{aq}, C = 0.001) \parallel \text{Ag}^+(\text{aq}, C = 0.1)/\text{Ag(s)}$
 Given: E° for $\text{Zn}^{2+}/\text{Zn} = -0.763$; E° for $\text{Ag}/\text{Ag}^+ = -0.80$

(OR)

14. (a) Derive the Nernst equation and calculate the E_{cell} value for the given cell: $\text{Zn}_{(s)}/\text{Zn}^{2+} (c = 0.01 \text{ M}) \parallel (c = 0.1 \text{ M}) \text{Cu}^{2+}/\text{Cu}_{(s)}$.
 Given: $E^0_{\text{Zn}/\text{Zn}^{2+}} = +0.76 \text{ V}$; $E^0_{\text{Cu}^{2+}/\text{Cu}} = +0.34 \text{ V}$. (10)
- (b) Explain the electrodeposition method with a suitable diagram. (5)
15. (a) How can we explain different physical properties of (i) H_2O and H_2S and (ii) Polyethylene and Teflon using various intermolecular force of interaction? (7)
- (b) Mention the difference between Viscosity & Viscosity Index in the context of lubricants. How do we determine viscosity using Redwood viscometer – elaborate with a suitable diagram? (8)

(OR)

16. (a) Define Flash point, Fire point, Cloud point and Pour point. What are the applications of these characteristics for a lubricant? (8)
- (b) Define acid value, saponification value and stem emulsification number for a lubricant with their importance. (7)
17. (a) Provide details classification of composite materials with suitable example. (8)
- (b) Provide the methods which are used for the synthesis of nanomaterial (bottom up). (7)

(OR)

18. (a) Why the nano particles have different reactivity compared to their corresponding bulk material. (3)
- (b) What is the role of stabilizer in nano particles synthesis – explain with an example. (2)
- (c) Write a short note on chemical vapor deposition and TiO_2 based self-cleaning glass. (5+5)

PART - C

Answer the following

1 x 20 = 20 Marks

19. (a) A sample of water on analysis was found to contain $\text{Ca}(\text{HCO}_3)_2 = 10.5 \text{ ppm}$; $\text{Mg}(\text{HCO}_3)_2 = 12.5 \text{ ppm}$; $\text{CaSO}_4 = 7.5 \text{ ppm}$; $\text{MgSO}_4 = 2.6 \text{ ppm}$. Calculate the temporary, permanent and total hardness of water in ppm. How much amount of 1/100 M EDTA solution is required for the titration of the 100 ml of the sample to determine the total hardness? [Mol. Wt. of $\text{Ca}(\text{HCO}_3)_2 = 162 \text{ g/mole}$, $\text{Mg}(\text{HCO}_3)_2 = 146 \text{ g/mole}$; $\text{CaSO}_4 = 136 \text{ g/mole}$; $\text{MgSO}_4 = 120 \text{ g/mole}$]. (5+3)
- (b) Compare the factors which influence corrosion. (4)
- (c) How does Iron undergo corrosion under acidic and basic conditions? Explain with suitable reaction scheme. (5)
- (d) Write down the working principle of DNA-directed self-assembly of AuNPs. (3)

**